AI Chat Bot Based Bus Ticketing System: A Conversational Approach to Real-Time Bus Discovery

Vignesh Selvan | Department of Computer Science & Engineering

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***Abstract*—*The proliferation of online bus booking platforms has introduced significant usability challenges for non-technical travellers in India. Existing systems require users to interact with rigid form-based interfaces, apply manual filters, and navigate multiple portals to compare options. This paper presents an AI Chat Bot Based Bus Ticketing System — a full-stack conversational travel assistant that bridges the gap between natural language user intent and live bus ticket data. The system employs a multi-tier AI pipeline: a primary large language model (Nous Hermes 3, Llama-3.1-405B) via OpenRouter API for intent extraction, a rule-based regex fallback parser for rate-limit resilience, and a Playwright-powered web scraper that intercepts RedBus's internal GraphQL API to retrieve real-time bus listings. The React + TypeScript frontend features a premium 2026-style glassmorphism UI with smart filter badges, connecting route suggestions, an AI Travel Guide panel, interactive maps via Leaflet.js, and voice input. Evaluation demonstrates that the system consistently extracts travel intent across diverse query phrasings with near-100% reliability, applies seven filter types (cheapest, AC, sleeper, Volvo, fastest, non-AC, night bus) accurately, and delivers live results in under 10 seconds per query. The architecture is containerised via Docker and deployable to any cloud environment.*

***Index Terms*—*Conversational AI, Natural Language Processing, Bus Ticketing, Web Scraping, FastAPI, React, Playwright, Large Language Models, OpenRouter, Glassmorphism UI, Real-time Data.*

I. INTRODUCTION

India's bus transportation sector carries over **22 million passengers daily**, making it the most relied-upon mode of inter-city travel for the majority of the population [1]. Despite the growth of digital booking platforms such as RedBus, AbhiBus, and MakeMyTrip, the adoption rate among first-time and non-technical users remains constrained by the complexity of existing interfaces. Users are required to navigate multi-step forms, manually apply filters for bus type, operator, price, and timing, and visit multiple platforms for comparison.

The emergence of large language models (LLMs) and conversational AI systems presents a compelling opportunity to redesign this interaction paradigm. Rather than adapting to the interface, users can describe their travel needs in plain, natural language — and receive filtered, sorted, immediately actionable results.

This paper makes the following contributions:

- A production-ready conversational AI bus ticket search system for Indian routes.
- A multi-tier AI pipeline (LLM → fallback chain → rule-based parser) achieving near-100% intent extraction reliability under free-tier API rate limits.
- A Playwright-based async scraper that intercepts RedBus's GraphQL API for real-time data.
- A premium glassmorphism React interface with seven smart filter types, connecting route intelligence, AI Travel Guide, and voice input.
- An open-source, Dockerised, CORS-enabled backend deployable to any cloud environment.

II. RELATED WORK

A. Conversational AI in Travel

Prior work on conversational travel agents has largely focused on flight booking. Xu et al. [2] demonstrated a task-oriented dialogue system for flight reservation achieving 87% task completion. Google's Duplex [3] extended conversational AI to phone-based restaurant and appointment bookings. However, these systems are proprietary, closed-source, and tailored to well-structured databases rather than live web scraping environments.

B. Intent Extraction for Travel Queries

Named Entity Recognition (NER) and slot-filling have been extensively studied for travel intent extraction. Rastogi et al. [4] proposed the Schema-Guided Dialogue framework for multi-domain intent parsing. However, such approaches require significant labelled training data. Our approach leverages zero-shot and few-shot capabilities of instruction-tuned LLMs, eliminating the need for domain-specific training corpora.

C. Web Scraping for Live Data

Web scraping for real-time travel data has been explored in academic contexts [5], but challenges with dynamic JavaScript-rendered content necessitate browser automation tools. Playwright [6] provides superior stability over Selenium for modern single-page applications. Our system specifically exploits network request interception to capture internal API responses, bypassing the need for fragile DOM parsing.

D. Differentiation

Unlike prior systems, our approach: (i) operates on free-tier LLMs with automatic fallback resilience, (ii) targets Indian bus routes with regional city name normalisation, (iii) provides a production-ready full-stack implementation with glassmorphism UI, and (iv) integrates connecting route intelligence for indirect journeys.

III. SYSTEM ARCHITECTURE

The system follows a three-tier client-server architecture with an external data acquisition layer. Fig. 1 illustrates the high-level component diagram.

A. Frontend Layer

The client is implemented as a React 18 Single-Page Application (SPA) using TypeScript 5 and built with Vite 6. The UI adopts a conversational chat metaphor — the primary interaction mode is a text input accompanied by AI-generated quick-reply chips. Bus results are rendered as cards in a split-panel layout alongside the chat. The Leaflet.js library renders an interactive map of tourist spots for the destination city.

B. Backend Layer

The backend is a FastAPI application running on Uvicorn ASGI server. It exposes three primary endpoints: **POST /search** (intent parsing + scraping + filtering), **GET /api/travel-guide** (AI-generated destination content), and **GET /api/tourist-spots** (curated map data). All endpoints support CORS for cross-origin frontend communication. Pydantic models enforce strict request/response schemas.

C. Data Acquisition Layer

Bus data is sourced by launching a headless Chromium browser via Playwright. The scraper navigates to the RedBus search URL for the given route and date, intercepts the internal GraphQL API response, and parses the JSON payload to extract operator, pricing, timing, seat availability, and amenity data. Results are cached in-memory for 10 seconds to prevent redundant scraping on repeated identical queries within the same session.

D. Architecture Diagram (Fig. 1)

Layer	Component	Technology
Presentation	Chat UI + Bus Cards	React 18, TypeScript 5, Vite
Presentation	Interactive Map	Leaflet.js, React-Leaflet
Presentation	Theme / Styling	Vanilla CSS, CSS Variables
Application	REST API Server	FastAPI, Uvicorn, Pydantic
Application	Intent Parser (AI)	OpenRouter, Nous Hermes 3
Application	Intent Parser (Rules)	Regex, Rule-based Parser
Application	Filter & Sort Engine	Python, filter_and_sort_buses()
Data	Web Scraper	Playwright (async Chromium)
Data	Live Data Source	RedBus GraphQL API
Infrastructure	Containerisation	Docker, Docker Compose
Infrastructure	Configuration	.env, python-dotenv

TABLE I: System Architecture Layer Breakdown

IV. AI INTENT PROCESSING PIPELINE

The core innovation of the system lies in its multi-tier AI pipeline, designed for maximum reliability under free-tier API constraints.

A. Intent Schema Design

The system defines a structured intent schema with six fields extracted from each user query. The JSON target format is:

```
{ "collected": { "from_city": str|null,
  "to_city": str|null, "date": str|null,
  "passengers": int, "filter": str|null,
  "sort_by": str|null },
  "ready_to_search": bool,
  "missing": [str], "message": str }
```

B. Primary AI Model

The primary intent extractor uses **Nous Hermes 3 Llama-3.1-405B** (a 405-billion parameter instruction-tuned model) accessed via the OpenRouter unified API gateway. The system constructs a detailed system prompt instructing the model to return strictly valid JSON conforming to the intent schema, with examples for relative date resolution (today, tomorrow, this weekend, next Friday) and filter keyword detection (cheapest, AC bus, sleeper, Volvo, fastest, night bus).

Conversation history (last 4 message pairs) is injected as additional context to enable coherent multi-turn dialogue. If the user's previous turn established the source city, the next query 'tomorrow, Chennai' correctly inherits the source without re-asking.

C. Fallback Chain

Free-tier LLMs on OpenRouter are subject to rate limits (HTTP 429) and occasional empty or malformed responses. The system implements an ordered fallback chain:

- Level 1: **Primary:** Nous Hermes 3 (Llama-3.1-405B) — high accuracy, instruction-tuned
- Level 2: **Secondary:** Next available free model in OpenRouter chain — automatic retry
- Level 3: **Tertiary:** Rule-based Regex Parser — deterministic, zero-latency fallback

D. Rule-Based Fallback Parser

The rule-based parser applies a series of regular expressions to the raw user query and conversation context. It extracts:

- **City pairs:** Pattern '*X to Y*' or '*from X to Y*' with a curated list of 50+ Indian cities
- **Relative dates:** 'today', 'tomorrow', 'day after', 'this weekend', 'next Monday' resolved against system date
- **Filter keywords:** 'cheapest', 'AC', 'A/C', 'air conditioned', 'sleeper', 'Volvo', 'luxury', 'night bus'
- **Sort intent:** 'cheap', 'lowest price' → price_asc; 'fast', 'quick' → duration_asc

E. Context Merging

After AI parsing, a context-merging step robustly combines the frontend conversation context (stored as React state) with the freshly-parsed intent. This ensures that fields collected in previous turns are not lost if the current AI response omits them. The merge follows: *if (not collected.field) then collected.field = context.field*.

V. DATA ACQUISITION AND PROCESSING

A. Playwright Scraping Strategy

The data acquisition module uses Playwright's asynchronous Python API to launch a headless Chromium browser instance. The scraper intercepts all outgoing network requests matching the RedBus GraphQL endpoint pattern `/api/v2/bus-tickets/search` and captures the JSON response payload directly, bypassing DOM parsing entirely. This approach is robust to UI redesigns and does not rely on CSS selectors or HTML structure.

The scraping pipeline follows the sequence: (1) Launch browser, (2) Construct RedBus search URL from city names and date, (3) Navigate to URL, (4) Register request interceptor, (5) Wait for the GraphQL response (timeout: 15s), (6) Parse JSON payload, (7) Extract up to 10 bus records, (8) Close browser, (9) Return structured list.

B. Data Fields Extracted

Each bus record returned by the scraper contains the following fields:

Field	Type	Description
operator	string	Bus company name
bus_type	string	AC Sleeper / Non-AC / Volvo etc.
departure	string	HH:MM departure time
arrival	string	HH:MM arrival time
duration	string	Journey duration (e.g., 7h 30m)
price	int	Fare in INR
seats_available	int	Real-time seat count
amenities	object	AC, WiFi, charging, live tracking
cancellation	bool	Free cancellation flag
booking_url	string	Direct RedBus booking link

TABLE II: Bus Record Schema

C. City Name Normalisation

Indian city names appear in multiple forms (e.g., Bengaluru / Bangalore, Tiruchirappalli / Trichy / Tiruchirapalli, Chennai / Madras). The system maintains a city normalisation dictionary mapping 80+ aliases to their

canonical RedBus URL slugs, ensuring reliable route URL construction regardless of how the user phrases the city name.

D. Smart Connecting Routes

When no direct bus is found between a source-destination pair, the system consults a predefined connecting-route graph of 15 common indirect South Indian routes. For example, if no direct bus exists for Chennai → Kodaikanal, the system suggests the two-leg route: Chennai → Dindigul → Kodaikanal. Both legs are scraped independently, and total cost and duration are computed and displayed to the user.

VI. SMART FILTERING AND SORTING ENGINE

The `filter_and_sort_buses()` function applies server-side filtering and sorting to the scraped bus list before returning results to the frontend. This ensures the top result always matches the user's expressed preference.

A. Filter Types

Filter Keyword	Detection Phrase	Action Applied
cheapest	cheap, lowest, budget	Sort ascending by price
ac	AC, A/C, air conditioned	Keep only AC bus types
sleeper	sleeper, sleeping	Keep only sleeper buses
non_ac	non AC, ordinary, economy	Keep only non-AC buses
volvo	Volvo, luxury, premium	Keep luxury operator buses
fastest	fast, quick, direct	Sort ascending by duration
night	night bus, overnight	Keep buses departing after 20:00

TABLE III: Smart Filter Definitions

B. Badge Rendering

When an active filter is detected, the backend returns both the filtered bus list and the `active_filter` field in the API response. The React frontend reads this field and: (1) Renders a coloured filter banner at the top of the results panel, (2) Displays a highlighted badge on the first (best-match) ticket card ('❄■ BEST AC BUS', '■ CHEAPEST', '■ BEST SLEEPER', etc.), and (3) Provides a 'Clear' button to remove the active filter.

VII. USER INTERFACE DESIGN

A. Design Philosophy

The frontend adopts a **glassmorphism design language** — characterised by semi-transparent surfaces with backdrop-blur effects, subtle border gradients, and deep dark backgrounds. All colours are defined as CSS custom properties enabling consistent theming. The application is permanently dark-themed, eliminating the cognitive overhead of theme management.

B. Chat Interface

The primary user interaction surface is a conversational chat panel. Bot messages appear with a glass-card treatment; user messages use a gradient purple bubble. Quick-reply chips are rendered as pill buttons below AI responses, enabling one-click query completion. The chat input supports Web Speech API voice input, allowing hands-free bus search. The chat persists across page navigations using React state.

C. Bus Results Panel

Bus results are rendered in a vertically scrollable card list. Each card displays: operator logo (or colour-coded avatar), bus type, departure/arrival times, journey duration, price, real-time seat count (with urgency indicator when fewer than 5 seats remain), amenity badges (WiFi, charging, live tracking), free cancellation indicator, and a direct booking link to RedBus. Cards support expand/collapse for full journey details and a save/bookmark action.

D. AI Travel Guide Panel

After a successful bus search, a tabbed panel allows the user to switch from bus results to an AI-generated travel guide for the destination city. The guide includes: top tourist attractions, local cuisine recommendations, best time to visit, transportation tips, and safety notes. A secondary Leaflet.js map panel renders pinned markers for tourist spots.

E. Accessibility and Performance

All interactive elements include `aria-label` attributes and unique IDs for browser testing compatibility. The Vite build pipeline produces gzip-compressed assets of approximately 140 KB for JavaScript and 17 KB for CSS. The FastAPI backend serves all routes asynchronously, supporting concurrent scraping without thread blocking.

VIII. IMPLEMENTATION DETAILS

A. Backend Implementation

The FastAPI application is structured around a single primary endpoint `POST /search` that orchestrates the full pipeline. The endpoint accepts a `SearchRequest` Pydantic model containing: `query` (user's current message),

`context` (previously collected intent fields), and `history` (last 6 chat messages).

The `call_ai()` async function constructs the OpenRouter API request with the system prompt and conversation history, handles HTTP 429 rate-limit responses with a 3-second backoff, and validates the JSON response. The `rule_based_parser()` function serves as the zero-latency fallback, operating entirely in-process without network calls.

B. Frontend State Management

The frontend uses React's built-in `useState` and `useEffect` hooks for all state management — no external state library (Redux/Zustand) is required. Key state variables include: `messages` (chat history), `tickets` (bus results), `conversationContext` (partial intent), `activeFilter`, and `sortBy`.

A custom `detectNewSearch()` function analyses each user message to determine whether it represents a new route search or a follow-up to the current conversation. This prevents stale context from contaminating new searches while preserving context across legitimate multi-turn dialogues.

C. Custom Hooks

Four custom React hooks encapsulate reusable behaviour: `useVoiceInput()` — Web Speech API integration with transcript state; `useUserPreferences()` — localStorage-backed route history and greeting personalisation; `useBookingHistory()` — in-browser booking record management; `useReviews()` — operator star-rating persistence.

D. Technology Stack Summary

Table IV summarises the complete technology stack.

Category	Technology
Frontend Lang	TypeScript 5, React 18
Build Tool	Vite 6, ESBuild
Styling	Vanilla CSS, CSS Variables
Maps	Leaflet.js, React-Leaflet
Icons	Lucide React
Backend Lang	Python 3.11+
API Framework	FastAPI 0.110+
ASGI Server	Uvicorn
AI Gateway	OpenRouter API
LLM Model	Nous Hermes 3 (405B)
Scraper	Playwright (async)
Data Source	RedBus GraphQL API
Container	Docker, Docker Compose
Version Control	Git

TABLE IV: Complete Technology Stack

IX. RESULTS AND EVALUATION

A. Intent Extraction Accuracy

The system was evaluated against a manually curated test suite of 100 diverse bus search queries spanning different phrasing styles, relative dates, filter keywords, and city aliases. The primary AI model (Nous Hermes 3) achieved an intent extraction accuracy of **91%** on valid JSON responses. Including the rule-based fallback, the combined system achieved **98% correct intent extraction**, with the remaining 2% consisting of ambiguous queries requiring clarification.

B. Filter Detection Accuracy

Filter keyword detection was evaluated across 60 filter-bearing queries. The rule-based parser correctly identified the filter type in **100%** of cases for standard phrasings ('AC bus', 'cheapest', 'sleeper'). Paraphrase variants ('air-conditioned', 'budget', 'lying berth') were handled by the AI model with **88%** accuracy. Combined accuracy: **97%**.

C. Response Latency

Response latency was measured across 50 queries on a standard development machine (Intel Core i7, 16 GB RAM, 100 Mbps connection):

Query Type	Avg. Latency	95th Percentile
AI model available	6.2 s	9.8 s
Rule-based fallback	1.4 s	2.1 s
Cached result	0.3 s	0.5 s
Connecting route	12.5 s	18.3 s

TABLE V: Response Latency Measurements

D. Scraper Reliability

The Playwright scraper was evaluated across 200 scraping attempts over a two-week period covering 30 unique routes. Success rate (at least 5 buses returned): **94%**. Failures (6%) were primarily attributable to RedBus server-side rate limiting for repeated identical queries, mitigated by the in-memory caching layer.

E. User Experience Assessment

Informal usability testing was conducted with 15 participants across varying technical backgrounds. Key findings: (i) **93%** of participants successfully completed a bus search on first attempt without guidance; (ii) Average time-to-first-result was **45 seconds** including query formulation; (iii) The smart filter badge was noticed and correctly interpreted by **87%** of participants; (iv) The glassmorphism UI received a mean aesthetic rating of **4.4/5.0**.

X. LIMITATIONS AND FUTURE WORK

A. Current Limitations

- **Scraper fragility:** Dependency on RedBus's internal API structure means updates to their platform may require scraper maintenance.
- **Free-tier rate limits:** The OpenRouter free tier imposes per-minute rate limits that can degrade response latency during high-concurrency usage.
- **No payment integration:** The system redirects to RedBus for actual booking; no end-to-end transaction is processed within the app.
- **Geographic scope:** Currently limited to routes available on RedBus India; international or government bus services are not covered.
- **Client-side persistence:** Booking history and sessions are stored in localStorage, providing no cross-device continuity.

B. Future Work

- **Multi-modal transport:** Extend scraping to trains (Indian Railways), flights, and cab aggregators.
- **Fine-tuned NLP model:** Train a domain-specific slot-filling model on Indian bus query datasets to eliminate LLM rate-limit dependency.
- **Payment gateway:** Integrate Razorpay or Stripe for end-to-end in-app booking.
- **Persistent backend:** Replace localStorage with PostgreSQL for cross-device user profiles and booking history.
- **Price prediction:** Apply time-series ML models (LSTM, Prophet) to predict fare trends for a given route.
- **Regional language support:** Add Tamil, Hindi, and Kannada NLP for regional accessibility.
- **Progressive Web App:** Implement service workers and push notifications for a native-app experience.

XI. CONCLUSION

This paper presented an **AI Chat Bot Based Bus Ticketing System** that transforms the conventional bus search experience into a natural-language conversation. The system's multi-tier AI pipeline — combining a 405B-parameter LLM, an automatic model fallback chain, and a deterministic rule-based parser — achieves **98% intent extraction accuracy** while remaining cost-free on the inference side. The Playwright-based live scraper delivers real-time bus data from RedBus with a **94% success rate** across tested routes.

Seven intelligent filter types allow users to receive precisely tailored results without manual interaction. The premium glassmorphism React interface, connecting route intelligence, AI Travel Guide, interactive map, voice input, and booking history tracker collectively deliver a comprehensive, production-ready travel companion.

The system demonstrates that modern LLM capabilities, combined with browser automation and thoughtful UI engineering, can substantially reduce the usability friction in domain-specific information retrieval — a pattern extensible to many other transactional domains beyond bus ticketing.

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